

Roll No.

TCS-502

B. TECH. (CSE) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022

OPERATING SYSTEMS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What are essential services provided by an operating system ? Justify selection of services. 10 Marks (CO1)

OR

- (b) Explain Booting process of a system. Also explain why we need an operating system. 10 Marks (CO1)

2. (a) What do you understand by Process ? Explain various states of process with a suitable diagram. 10 Marks (CO1)

OR

- (b) Which data structure is used by an operating system to keep track of process information ? Explain. 10 Marks (CO1)

3. (a) Explain various types of schedulers and their importance in process management. 10 Marks (CO1)

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(2)

OR

- (b) Compare and contrast symmetric and asymmetric multiprocessing architecture. 10 Marks (CO1)

4. (a) Consider the following table : 10 Marks (CO2)

Process	Arrival Time	Burst Time	Priority
1	8	12	8
2	6	5	10
3	2	6	6
4	4	8	12

Determine the turnaround time and response time for each process and the average turnaround time for all jobs for the following scenario :

Dynamic priority scheduling where priority of running process will decrease by 2 after every 2 unit of running time. Here lower number in priority column represents higher priority.

OR

- (b) Four jobs are waiting to be run. Their expected run times are 6, 3, 5 and x. In what order should they be run to minimize the average response time ? Justify your answer. 10 Marks (CO2)

5. (a) What is a race condition ? Explain, how a critical section avoids this condition. What are the properties which a data item should possess to implement a critical section ? 10 Marks (CO2)

(3)

OR

(b) Suppose you try to enforce mutual exclusion as follows :

PROCESS 1	PROCESS 2
<pre>while (true) { while (p2_in_crit_sect == true) ; /* do nothing */ P1_in_crit_sect = true; ... critical section ... p1_in_crit_sect = false; }</pre>	<pre>while (true) { while (p1_in_crit_sect == true) ; /* do nothing */ P2_in_crit_sect = true; ... critical section ... p2_in_crit_sect = false; }</pre>

Does this code always enforce mutual exclusion ? Justify your answer.

10 Marks (CO2)